CUSTOMER SHOPPING BEHAVIOR ANALYSIS

1. PROJECT OVERVIEW

This project analyzes transactional data from 3,900 purchases across multiple product categories to uncover insights into customer purchasing behavior. The goal is to identify spending trends, segment customers, understand product preferences, and examine subscription patterns — all to support data-driven business decision-making.

2. DATASET SUMMARY

The dataset consists of 3,900 records across 18 columns, capturing a range of customer and transaction attributes. Customer-related fields include age, gender, location, and subscription status, while purchase details cover the item bought, its category, price, season, size, and color. Shopping behavior is further captured through variables such as discount and promo code usage, previous purchases, purchase frequency, review ratings, and shipping type. Notably, the Review Rating column contains 37 missing entries, which should be accounted for during analysis.

3. EXPLORATORY DATA ANALYSIS USING PYTHON

The analysis was performed with preparing and cleaning the dataset in Python:

- **Data Loading:** The dataset was imported into the environment using the pandas library.
- **Initial Exploration:** The data structure was reviewed using `df.info()` , and summary statistics were produced with `df.describe()` .
- **Handling Missing Data:** Null values were detected, and missing entries in the Review Rating column were filled using the median rating for each product category.
- **Column Standardization:** Column names were reformatted to snake_case to improve readability and ensure consistent documentation throughout.
- **Feature Engineering:**
    - An `age_group` column was introduced by grouping customer ages into defined bins.

- A `purchase_frequency_days` column was calculated from purchase history to reflect customer buying patterns.

- **Data Consistency Check:** A review of `discount_applied` and `promo_code_used` revealed significant overlap, leading to the removal of the `promo_code_used` column.
- **Database Integration:** The cleaned DataFrame was linked to a SQL database to support further SQL-based analysis.

4. DATA ANALYSIS USING SQL

1. Revenue by Gender: Compared total revenue generated by male and female customers to identify any gender-based spending trends.

Results | Messages

	gender	revenue
1	Male	$157,890.00
2	Female	$75,191.00
3	Total	$233,081.00

2. High-Spending Discount Users: Highlighted customers who applied discounts but still spent above the average purchase amount, indicating strong purchasing potential.

Results | Messages

	customer_id	purchase_amount
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91
16	37	69
17	40	60

Query executed successfully.

3. Top Products by Rating: Determined the products with the highest average review ratings to identify customer favorites.

Results | Messages

	item_purchased	avg_rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.8
5	Handbag	3.78

4. Shipping Type Comparison: Analyzed average purchase amounts for Standard versus Express shipping to understand customer preferences.

Results | Messages

	shipping_type	avg_purchase_amount
1	Standard	58
2	Express	60

5. Subscribers vs. Non-Subscribers: Compared average spend and total revenue across subscription status to assess the impact of subscriptions on revenue.

Results | Messages

	subscription_status	(No column name)	avg_spend	total_revenu
1	Yes	1053	59	62645
2	No	2847	59	170436

6. Discount-Dependent Products: Identified the five products with the highest proportion of purchases made with discounts.

Results | Messages

	item_purchased	discount_rate
1	Hat	50.000000000000
2	Sneakers	49.660000000000
3	Coat	49.070000000000
4	Sweater	48.170000000000
5	Pants	47.370000000000

7. Customer Segmentation: Classified customers into New, Returning, and Loyal segments based on their purchase history to enable targeted marketing strategies.

Results | Messages

	customer_segment	number_of_customers
1	New	83
2	Returning	701
3	Loyal	3116

8. Top Products per Category: Listed the three most purchased products within each product category to reveal popular items.

Results | Messages

	item_purchased	category	total_purchased	item_rank
1	Jewelry	Accessories	171	1
2	Belt	Accessories	161	2
3	Sunglasses	Accessories	161	2
4	Scarf	Accessories	157	3
5	Blouse	Clothing	171	1
6	Pants	Clothing	171	1
7	Shirt	Clothing	169	2
8	Dress	Clothing	166	3
9	Sandals	Footwear	160	1
10	Shoes	Footwear	150	2
11	Sneakers	Footwear	145	3
12	Jacket	Outerwear	163	1
13	Coat	Outerwear	161	2

9. Repeat Buyers & Subscriptions: Investigated whether customers with more than five purchases were more likely to have active subscriptions.

Results | Messages

	subscription_status	repeat_buyers
1	Yes	958
2	No	2518

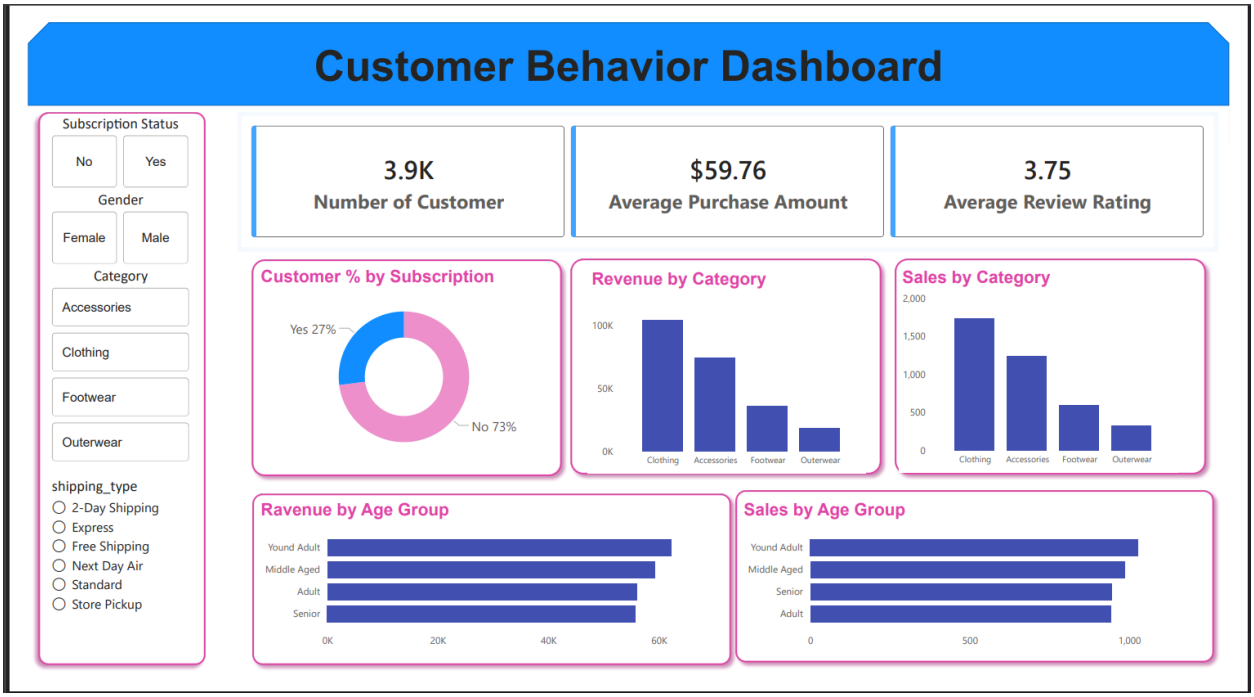
10. Revenue by Age Group: Calculated the total revenue contribution of each age group to identify the most valuable demographic segments.

Results | Messages

	age_group	total_revenue
1	Yound Adult	62143
2	Middle Aged	59197
3	Adult	55978
4	Senior	55763

5. DASHBOARD IN POWER BI

An interactive dashboard was created in Power BI to visually present the key insights and facilitate intuitive exploration of the data.



6. BUSINESS RECOMMENDATIONS

- Only 27% of customers currently have subscriptions. Offering exclusive deals to subscribers could encourage more sign-ups and help cultivate a larger base of loyal customers.
- Outwear has the lowest sales and revenue. Increasing marketing efforts for this category and designing an appealing display could help boost sales.
- Introducing discounts targeted at senior customers may encourage higher spending and contribute to increased revenue.
- Since there are currently no female subscribers, providing attractive promotions on products for women could help engage this segment and potentially convert them into subscribers.